Topic 2.11 - Logarithmic Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________ Class: ____________ Date: ____________

Why this topic matters

A logarithmic function compresses multiplicative input change into additive output change. Its vertical asymptote marks the boundary of the domain, and transformations must be interpreted through the argument as well as the outside coefficient.

Learning targets

- Describe logarithmic domain, range, asymptotes, monotonicity, concavity, and end behavior.
- Connect equal multiplicative changes in input to equal additive changes in output.
- Analyze and construct transformed logarithmic functions with explicit restrictions.

Language and structure

Term	Meaning in this topic
vertical asymptote	A vertical line approached as the logarithm argument tends to 0^+.
multiplicative interval	An input change by a constant factor rather than a constant difference.
anchor point	For $a\log_b(x-h)+k$, the point $(h+1,k)$ because $\log_b 1=0$.
monotonicity	Increasing or decreasing behavior determined by the base and any reflection.
concavity	For $\log_b x$, concave down when $b>1$ and concave up when $0<b<1$.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A function changes by $+3$ whenever its positive input is multiplied by 10. What family is suggested, and what information is still needed to determine a unique formula?

Concept 1: The parent logarithmic family

Core idea

For $f(x) = \log_b x$, the domain is $(0, \infty)$, the range is $\mathbb{R}$, and the vertical asymptote is $x = 0$.

- If $b > 1$, the function is increasing and concave down.
- If $0 < b < 1$, the function is decreasing and concave up.
- The graph passes through $(1, 0)$ and $(b, 1)$.

Worked example - annotate the reasoning

Problem. Analyze $f(x) = \log_3 x$: state domain, range, intercept, asymptote, monotonicity, concavity, and end behavior.
Model reasoning. Domain $(0, \infty)$, range $\mathbb{R}$, x-intercept $(1, 0)$, no y-intercept, asymptote $x = 0$, increasing and concave down. As $x \to 0^+$, $f(x) \to -\infty$; as $x \to \infty$, $f(x) \to \infty$.

Check your understanding

Independent

Analyze $g(x) = \log_{1/2} x$.

Concept 2: Transformations and domain

Core idea

For $f(x) = a \log_b(x - h) + k$, the domain is $x > h$, the asymptote is $x = h$, and the anchor point is $(h + 1, k)$.

- The sign of a can reverse monotonicity and concavity.
- Horizontal behavior comes from setting the argument equal to familiar powers of the base.
- Never state the domain from the outside shift; solve the argument inequality.

Worked example - annotate the reasoning

Problem. Analyze $f(x) = -3 \log_4(x - 2) + 5$.
Model reasoning. Domain $(2, \infty)$; range $\mathbb{R}$; asymptote $x = 2$; anchor point $(3, 5)$. Since $\log_4$ is increasing and concave down, multiplication by -3 makes f decreasing and concave up.

Check your understanding

Independent

For $p(x) = 2\ln(5 - x) - 1$, state the domain and asymptote.

Concept 3: Multiplicative input change

Core idea

For $f(x) = a\log_b x + k$, multiplying the input by b adds a to the output.

- $f(bx) - f(x) = a$.
- More generally, multiplying input by b^m changes output by am.
- This property explains why logarithmic scales convert ratios into differences.

Worked example - annotate the reasoning

Problem. For $f(x) = 4\log_2 x - 3$, compare $f(8x)$ with $f(x)$.
Model reasoning. $8 = 2^3$, so $f(8x) - f(x) = 4 \cdot 3 = 12$. Thus $f(8x) = f(x) + 12$.

Check your understanding

Independent

A logarithmic model increases by 6 whenever the input is multiplied by 5. Write one possible model.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

State the domain and asymptote of $f(x) = \log_7(x+4) - 2$.

Exit ticket 2

No calculator

Find the anchor point of $g(x) = 5\ln(x-3) + 1$.

Exit ticket 3

No calculator

For $h(x) = 2\log_3 x$, how does $h(9x)$ compare with $h(x)$?